

Francisco Girbal Eiras

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Github · Google Scholar

ABOUT ME	I am currently working on evaluations and benchmarks for compliance of generative AI systems under custom safety specifications, with a focus on LLMs and tool calling agents. Previously, I completed a PhD on Trustworthy Machine Learning at the University of Oxford under the supervision of Philip Torr, Adel Bibi and M. Pawan Kumar (Google DeepMind).	
INDUSTRY EXPERIENCE	Dynamo AI, ML Research Scientist	Nov. 2024 – Present
	- Leading the research team developing agentic red-team evaluations for LLMs, targeting custom safety specifications in conversational and tool-using agents. - Designing automated pipelines to increase adversarial coverage and behavioral diversity across evaluation scenarios. - Contributing to core research efforts, including a first-author paper on the robustness of LLM safety judges and a benchmarking hallucination paper currently under review.	
	Five, Research Scientist Intern	Dec. 2022 – Jun. 2023
	Worked on efficient methods to perform zero-shot and weakly-supervised referring image segmentation (i.e., segmenting an object in an image that is referred in a natural language sentence), achieving new state-of-the-art performance in the field.	
	Five, Research Scientist Intern	Jun. 2021 – Sep. 2021
	Extended the certified robustness technique of randomized smoothing from isotropic ℓ_p balls to <i>anisotropic</i> certificates through a simplified Lipschitz analysis-based framework.	
	Five, Research Engineer	Sep. 2018 – Sep. 2020
	- Led the development of safe and scalable optimization-based motion planning algorithms, working in a team with research scientists and software engineers. - Published and presented research work developed at top tier conferences and journals within the robotics community, as well as to non-technical audiences. - Wrote and reviewed research and development code, ensuring CI with other tools within the company.	
	Institute for Systems and Robotics, Graduate Research Assistant	Apr. 2017 – Sep. 2017
	Developed new methods to perform pose estimation using vanishing points in general (central and non-central) omnidirectional cameras leading to a CVPR 2018 paper.	
EDUCATION	University of Oxford, Oxford, UK <i>DPhil (PhD), Engineering Science, AIMS</i> Title: <i>Towards Trustworthy Machine Learning Models in Vision, Physics, and Language Applications</i> Supervisors: Prof. Philip H.S. Torr, Dr. Adel Bibi, Dr. M. Pawan Kumar (Google DeepMind)	Oct. 2020 – Oct. 2024
	University of Oxford, Oxford, UK <i>MSc, Computer Science</i> Grade: Distinction	Oct. 2017 – Sep. 2018
	Técnico Lisboa, Lisbon, Portugal <i>BSc, Electrical and Computer Engineering</i> GPA 18/20; top 2% of class	Sep. 2013 – Jul. 2016
SELECTED PUBLICATIONS	F Eiras , E. Zemour, E. Lin, V. Mugunthan, <i>Know Thy Judge: On the Robustness Meta-Evaluation of LLM Safety Judges</i> , International Conference on Learning Representations (ICLR) Workshop Proceedings, 2025	

F Eiras, A Petrov, PHS Torr, MP Kumar, A Bibi, *Do as I do (Safely): Mitigating Task-Specific Fine-tuning Risks in Large Language Models*, International Conference on Learning Representations (ICLR), 2025

F Eiras, R Bunel, K Dvijotham, A Bibi, PHS Torr, MP Kumar, *Efficient Error Certification for Physics-Informed Neural Networks*, International Conference on Machine Learning (ICML), 2024

F Eiras, A Petrov, B Vidgen, [...], T Darrell, Y Lee, J Foerster, *Near to Mid-term Risks and Opportunities of Open-Source Generative AI*, International Conference on Machine Learning (ICML, Position Paper), 2024 [Oral]

F Eiras, K Oksuz, A Bibi, PHS Torr, PK Dokania, *Segment, Select, Correct: a Framework for Weakly-Supervised Referring Segmentation*, European Conference on Computer Vision (ECCV) Workshop Proceedings, 2024

F Eiras, R Bunel, K Dvijotham, A Bibi, PHS Torr, MP Kumar, *Provably Correct Physics-Informed Neural Networks*, 2nd Workshop on Formal Verification of Machine Learning, International Conference on Machine Learning (ICML), 2023 [Outstanding Paper Award]

A Petrov, **F Eiras**, A Sanyal, PHS Torr, A Bibi, *Certifying Ensembles: A General Certification Theory with S -Lipschitzness*, International Conference on Machine Learning (ICML), 2023

F Eiras, M Alfara, MP Kumar, PHS Torr, PK Dokania, B Ghanem, A Bibi, *ANCER: Anisotropic certification via sample-wise volume maximization*, Transaction of Machine Learning Research (TMLR), 2022

F Eiras, M Hawasly, SV Albrecht, S Ramamoorthy, *A Two-Stage Optimization-based Motion Planner for Safe Urban Driving*, Transaction on Robotics (T-RO), 2021

SV Albrecht, C Brewitt, J Wilhelm, B Gyevnar, **F Eiras**, M Dobre, S Ramamoorthy, *Interpretable Goal-based Prediction and Planning for Autonomous Driving*, International Conference on Robotics and Automation (ICRA), 2021

HONORS & AWARDS

Outstanding Paper Award (by European Lighthouse on Secure and Safe AI), 2nd Workshop on Formal Verification of Machine Learning @ ICML, 2023

Honorable Mention – Using Computer Vision for Social Good, LauzHack, 2018

Graduate Research Fellowship, Institute for Systems and Robotics, 2017

Swiss-European Mobility Studentship, 2016

Undergraduate Research Fellowship, Institute for Systems and Robotics, 2016

Undergraduate Academic Excellency Award, Técnico Lisboa, 2013 – 2016

INVITED TALKS

On Fine-Tuning Risks in Closed Large Language Models

OxAI (Oxford AI Society) Mini-Conference, Feb. 2024

Towards Certified Machine Learning

Columbia University, New York University, University of California, Berkeley, Stanford University, Jul. 2023

Provably Correct Physics-Informed Neural Networks

UK AI Fellows Conference (Turing Institute), May 2023

RESEARCH INTERESTS

AI Agents

LLM Evals and Benchmarking

Certified Machine Learning

Optimization

Adversarial Robustness

Multimodal and Self-Supervised Learning

Formal Methods

PROGRAMMING/ FRAMEWORKS

Python

C/C++

PyTorch

Tensorflow

WandB

transformers

git

CI/CD

AWS

Docker

React

HTML+CSS

Javascript

flask

POV-Ray